MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 03** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal choice.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The area enclosed between two concentric circles of radii 7 cm and 14 cm is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 462 cm^2 (B) 308 cm^2 (C) 154 cm^2 (D) 616 cm^2

Q2. There are two concentric circles of radii 8 cm and 5 cm. A sector forms an angle of 60° at the common centre O. The area of the shaded sector annulus is: [1]

(A) $\frac{77}{2}\pi$ cm^2 (B) $\frac{13}{2}\pi$ cm^2 (C) $\frac{39}{4}\pi$ cm^2 (D) $\frac{65}{6}\pi$ cm^2

Q3. A wheel of diameter 42 cm makes 1000 revolutions. The total distance covered by the wheel is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 1.32 km (B) 13.2 km (C) 132 m (D) 1320 cm

Q4. The minute hand of a clock is 14 cm long. The area swept by the minute hand in 5 minutes is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 51.33 cm^2 (B) 154 cm^2 (C) 25.67 cm^2 (D) 77 cm^2

Q5. **Assertion (A):** The total perimeter of a semicircular protractor of radius r is $(\pi + 2)r$.
Reason (R): The perimeter of a semicircular region consists of the length of the semicircular arc plus the length of its bounding diameter. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Find the area of a quadrant of a circle whose circumference is 22 cm. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors. Find: (i) the total length of the silver wire required, and (ii) the area of each sector of the brooch. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

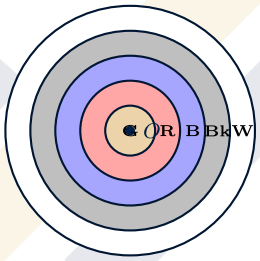
Q8. A horse is tied to a peg at one corner of a square grass field of side 15 m by a 5 m long rope. Find: (i) the area of that part of the field in which the horse can graze, and (ii) the increase in the grazing area if the rope were 10 m long instead of 5 m. (Use $\pi = 3.14$) [3]

Q9. To warn ships for underwater rocks, a coastal lighthouse spreads a red colored light over a sector of angle $80°$ to a distance of 16.5 km. Find the area of the sea over which the ships are warned. (Use $\pi = 3.14$) [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Olympic Archery Target Concentric Scoring Bands

An archery target has five concentric scoring zones from the centre outward: Gold, Red, Blue, Black, and White. The diameter of the central Gold zone is 21 cm, and each of the outer four rings has a uniform radial width of 10.5 cm.



(i) Determine the radius of the outer boundary of the Red scoring ring. [1]

(ii) Calculate the area of the central Gold scoring region. [1]

(iii) Calculate the surface area of the Red scoring band, OR show that the area of the Red band is exactly three times the area of the Gold region. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

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